

Session_06

Q.1 Write a program that prints the numbers from 10 down to 1 using a for loop, similar to a countdown timer.

```
A.    #include <stdio.h>
      int main() {
          int i;
          for (i = 10; i >= 1; i--)
          {
              printf("%d\n", i);
          }
          return 0;
      }
```

Q.2 Create a menu-driven console app that lets the user: 1) View your favorite 3 IPL teams, 2) Add a new team, 3) Exit. Use a while loop to keep showing the menu until the user chooses Exit. Use input() (or Scanner in Java) to get the user's choice each time.

```
A.2    #include <stdio.h>
      int main() {
          int choice;

          while (1)
          {
              printf("\n--- IPL Teams Menu ---\n");
              printf("1. View favorite 3 IPL teams\n");
              printf("2. Add a new team\n");
              printf("3. Exit\n");
              printf("Enter your choice: ");
              scanf("%d", &choice);

              if (choice == 1)
              {
                  printf("\nFavorite IPL Teams:\n");
                  printf("1. Mumbai Indians\n");
                  printf("2. Chennai Super Kings\n");
                  printf("3. Royal Challengers Bengaluru\n");
              }
              else if (choice == 2)
```

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```
{
    printf("Enter new team name: "); // Team can be entered here
    printf("New team added!\n");
}
else if (choice == 3)
{
    printf("Exiting the program...\n");
    break;
}
else {
    printf("Invalid choice! Please try again.\n");
}
}
return 0;
}
```

Q.3 Build a 'Guess the Song' game like Spotify — the program randomly picks a song name from a list and asks the user to guess it. Use a do-while loop so the user can keep guessing until they get it right. Use at least 3 song names of your choice.

```
A.3    #include <stdio.h>
        #include <string.h>

        int main() {
            char song[50];
            char guess[50];

            printf("Guess the Song!\n");
            printf("Hint: It is a popular song.\n");

            strcpy(song, "Perfect");

            do {
                printf("Enter your guess: ");
                scanf("%s", guess);
                if (strcmp(guess, song) == 0) {
                    printf("Correct! You guessed the song!\n");
                }
            }
        }
```

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```
else {  
printf("Wrong guess! Try again.\n");  
}  
} while (strcmp(guess, song) != 0);  
return 0; }
```

Q.4 Explain with your own example the difference between entry-controlled and exit-controlled loops by writing a short code snippet for each (for/while vs do-while) and describing what happens if the loop condition is false at the start.

A.4 Entry-controlled loop — while

```
#include <stdio.h>  
int main() {  
    int i = 10;  
    while (i < 5) {  
        printf("%d\n", i);  
        i++;  
    }  
    return 0;  
}
```

Exit-controlled loop — do-while

```
#include <stdio.h>  
int main() {  
    int i = 10;  
    do {  
        printf("%d\n", i);  
        i++;  
    } while (i < 5);  
    return 0;  
}
```
